4.3 Enterprise Data Warehouse

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OBJECTIVE

This exercise shows how to create an enterprise data warehouse. It illustrates how to create advanced InfoObjects such as an aDSO needed for loading transactional data and a composite provider needed for reporting.

ACTIVITIES

- Learn how to create an advanced DataStore Object and how to load it with data.
- Learn how to create the virtualization layer of the data warehouse.

SOFTWARE PREREQUISITES

- Eclipse with SAP BW Modeling Tools
- SAP GUI

DATA SET

Global Bike (GB) data set

SYSTEM ACCESS

SAP BW/4HANA 2.0 from your UCC

Scenario

Jan has already loaded the transaction data from the source system into the data acquisition layer (exercise 4.2), modelled InfoObjects as elementary information objects (exercise 3.2) and loaded the master data (exercise 4.1). In this exercise Jan will model the data warehouse and data mart layer and load the transaction data from the data acquisition layer.

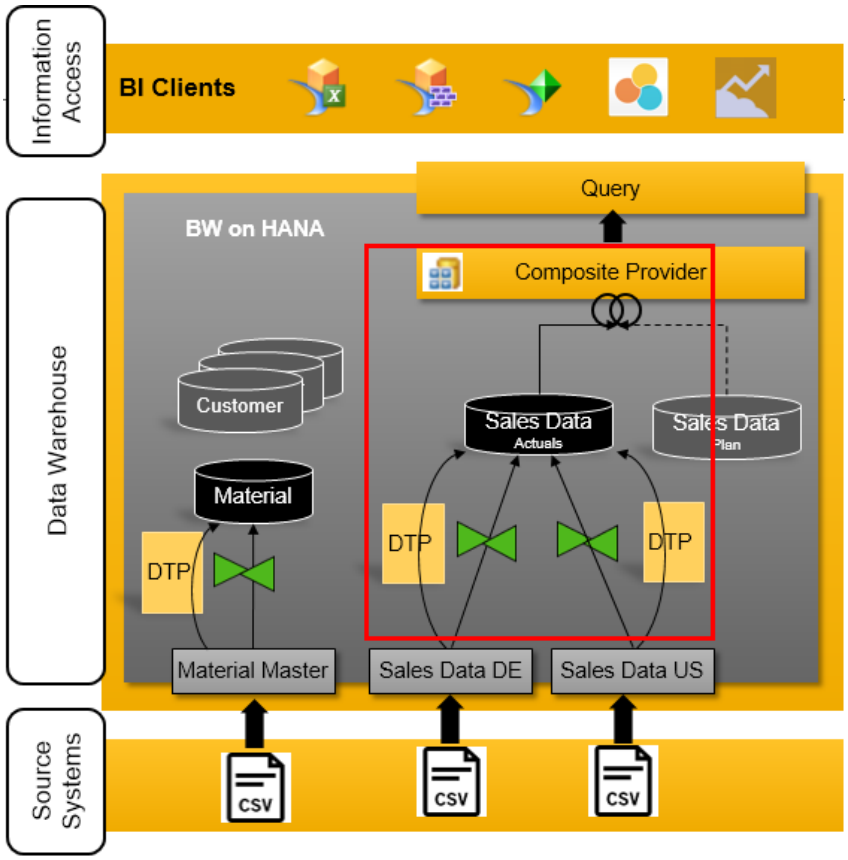


Figure 1

In a previous exercise (2) the sales transaction data were stored in the data acquisition layer. In the next steps it will be loaded into the data warehouse. First you need an advanced DataStore Object (aDSO) based on the sales transaction data DataSource. Then you will load the sales transaction data from Global Bike Germany and US in the data warehouse. Finally you will create a Composite Provider as a virtualization layer for analytics.

The graphic below displays the complete process (8 tasks).

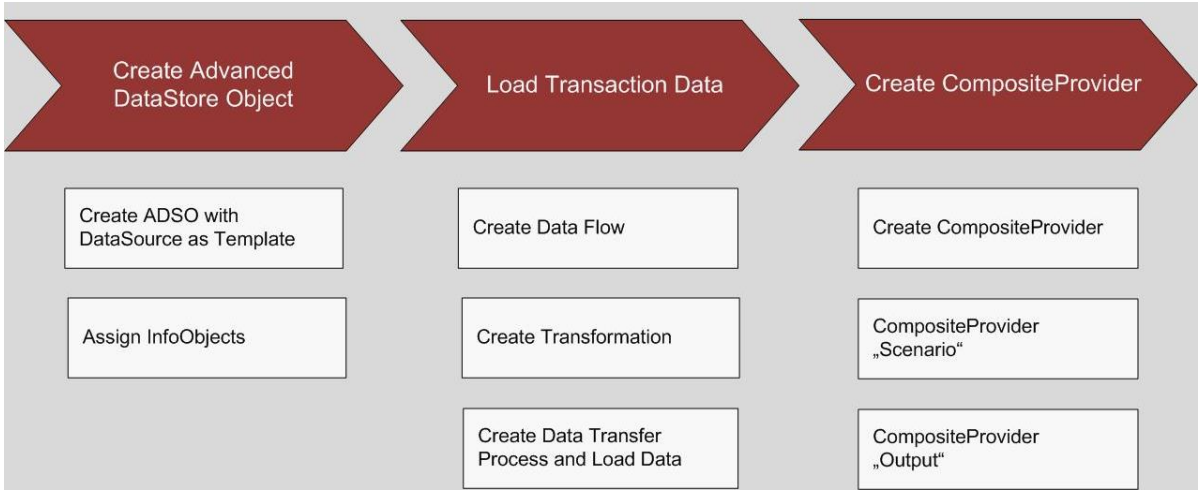


Figure 2

1. Create aDSO with DataSource as a Template

Jan uses Eclipse to create an *Advanced DataStore Object (aDSO)* for sales transaction data of Global Bike Germany and US.

1. Start Eclipse and open your BW project: **Project Explorer → your BW-System → BW Repository**.

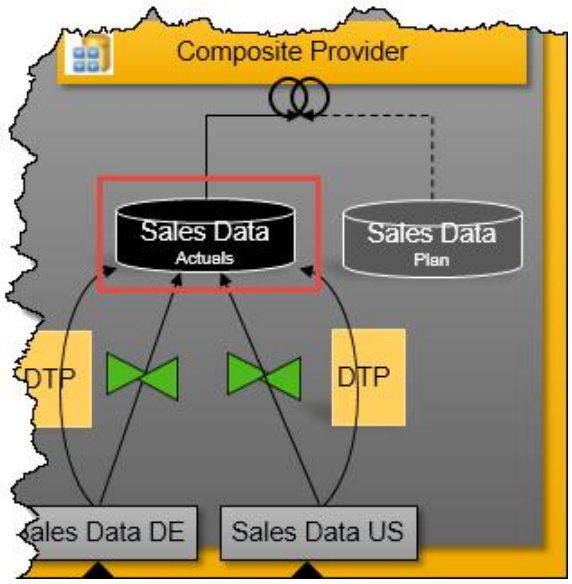


Figure 3

2. Log on and navigate to the InfoArea assigned by your instructor, which has the technical name XXX### and description Student ###.
3. Right-click and choose **New → DataStore Object (advanced)**.

Note: When working with the exercise, please replace in all names XXX with the Id of your course, and ### with your user number (three digits) provided by your instructor.

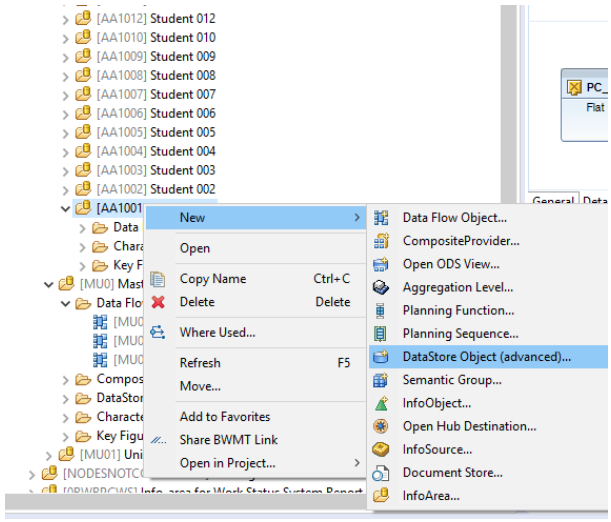


Figure 4

4. On the following screen, name XXXDM1### and description XXX### Sales Data aDSO. Use one of your DataSources from exercise 4.2 Data Acquisition as a template (e.g. XXXDSDE###).
5. All fields of your DataSource are copied as fields to DataStore Object (includes type information and other metadata).

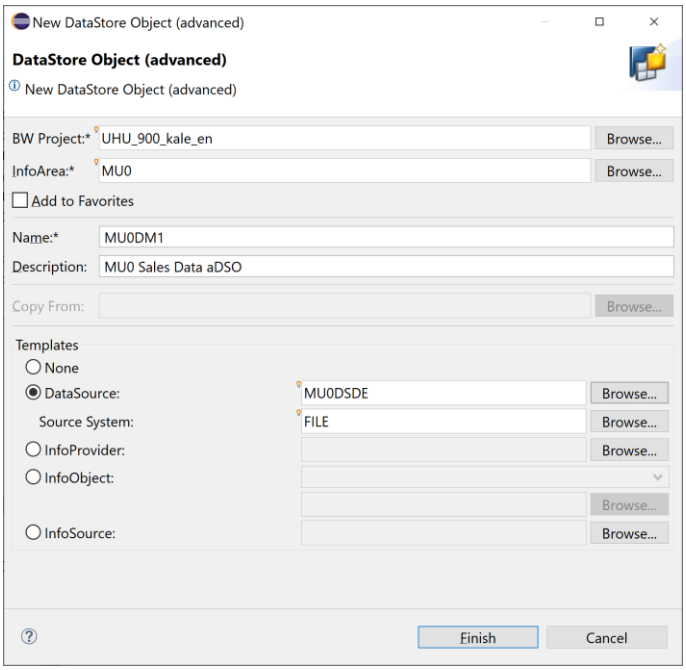


Figure 5

6. **Finish.**
7. On **General** tab, check **Data Mart DataStore Object** option. This sets some general properties in the aDSO, like activation of data after updates or key fields of the aDSO.

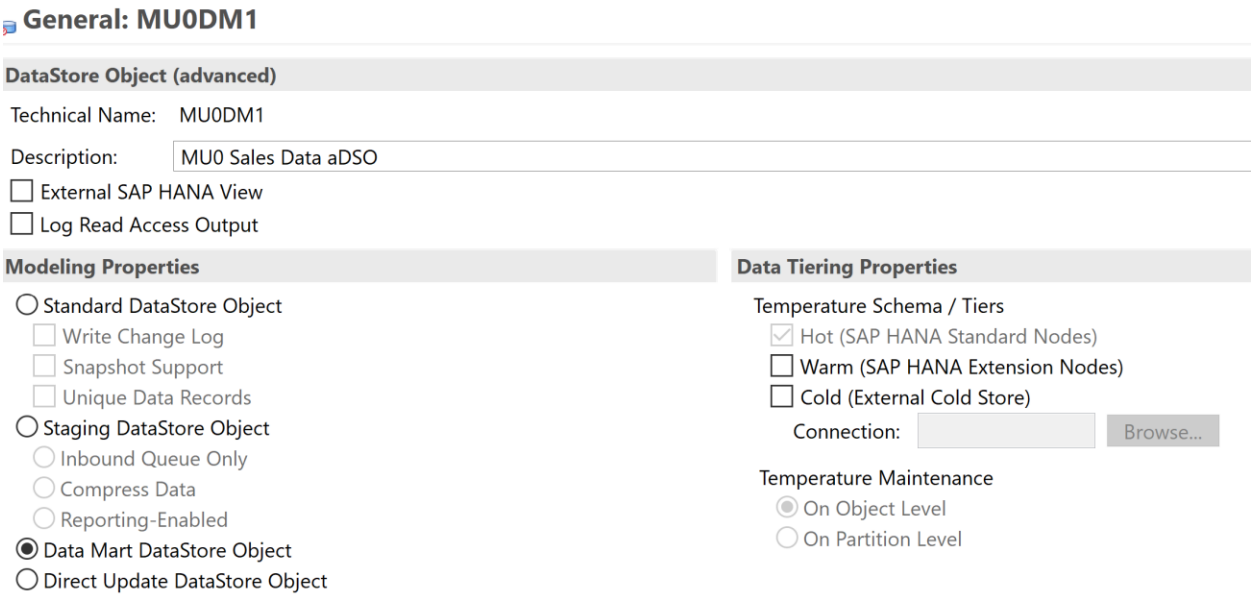


Figure 6

8. **Save** your aDSO.

2. Assign various InfoObjects to your Advanced DataStore Object.

Jan uses Eclipse to assign various InfoObjects to the aDSO.

1. Start **Eclipse** and open your BW project: **Project Explorer → your BW System → BW Repository**.
2. If you closed Eclipse, log on again and navigate to the InfoArea assigned by your instructor, which has the technical name XXX### and description Student ###.
3. Open your aDSO from the previous step (technical name: XXXDM1###).
4. On **Details** tab you see a list of all fields that have been copied from the DataSource template. On this tab, we will model the following:
   a. **Use InfoObjects instead of fields**: this step for creating valuable information out of data ("data to information"), as the data source is just the inbound representation of an external source of data, whereas InfoObjects represent company wide, integrated, and harmonized information objects.
   b. **Add InfoObjects** that enrich our aDSO with information that were not a part of the original.

5. We will group the fields/InfoObjects into groups for a better overview. The groups correspond to dimensions in a multidimensional model.
6. First, create 5 groups: **ORDER**, **CUSTOMER**, **PRODUCT**, **TIME**, and **MEASURES**.

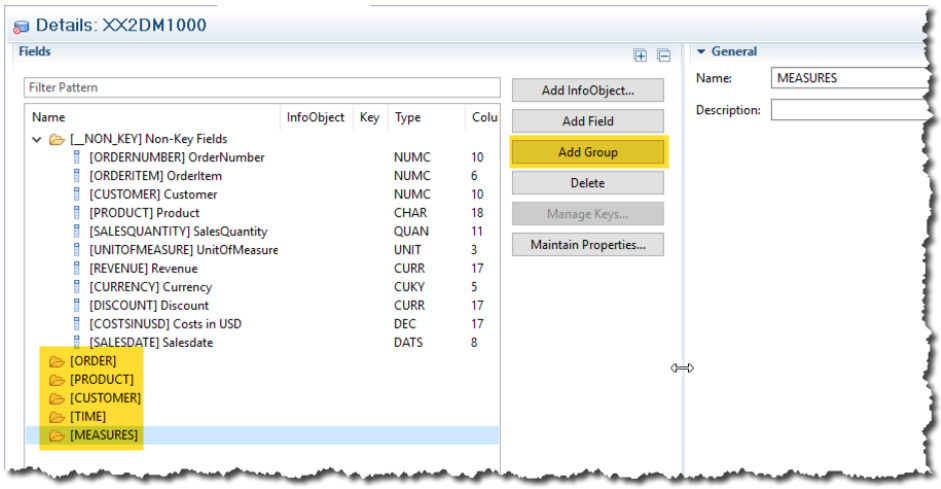


Figure 7

7. **Save**.
8. Then move the fields ORDERNUMBER & ORDERITEM into **ORDER**. For these fields, there are no corresponding InfoObjects in the system. *Fields have a different icon than InfoObjects. Pay attention to the difference.*
9. In the following steps, you will change fields into corresponding InfoObjects.

Hint: Make use of the naming convention to find out the technical names of the corresponding InfoObjects.

- *All InfoObjects delivered by SAP have a name starting with **0**....*
- *All InfoObjects delivered by UCC have a name starting with **MU0**...*
- *All InfoObjects, you created have a name starting with your **XXX**...*

10. Click on the Customer field. Then click **Change Type** and then browse for MU0CUST. This is how it should change from Field to InfoObject. Note its icon.
11. Then use the following table to change fields into InfoObjects.

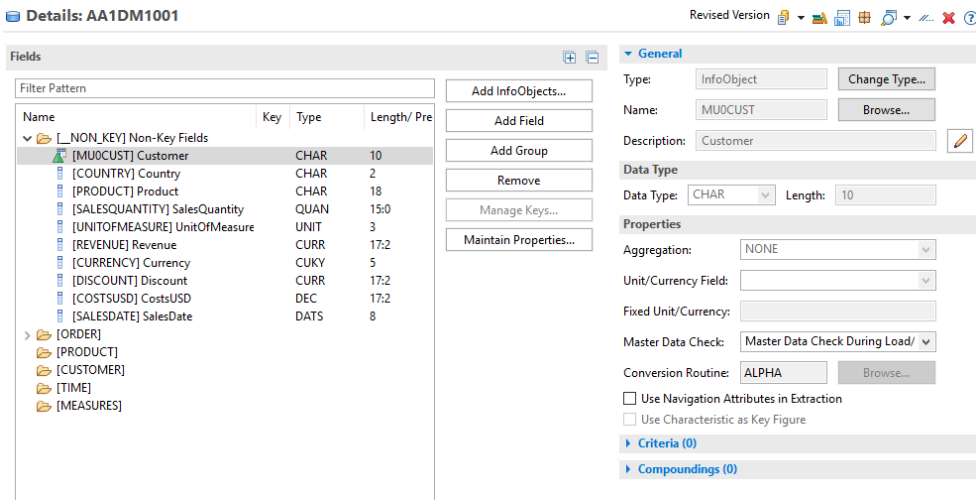


Figure 8

Group	Field	Change Field to the following InfoObject
CUSTOMER	Customer	Customer (delivered by UCC)
PRODUCT	Product	Material (created by you)
TIME	Salesdate	Calendar Day (delivered by SAP)
MEASURES	UnitofMeasure	0UNIT
MEASURES	SalesQuantity	Sales Quantity (created by you)
MEASURES	Revenue	Revenue (created by you)
MEASURES	Discount	Discount (delivered by UCC)
MEASURES	CostsInUSD	Cost of Goods manufactured USD (delivered by UCC)

12. Now **move** the InfoObjects into the corresponding group according to the following table.

GROUP	Name	Technical Name
Non-Key Fields	Unit of Measure	UNITOFMEASURE
	Currency	CURRENCY
ORDER	OrderItem	ORDERITEM
	OrderNumber	ORDERNUMBER
PRODUCT	Material	XXXMA###
CUSTOMER	Customer	MU0CUST
TIME	Calendar Day	0CALDAY

MEASURE	CostUSD	MU0COGM
	Discount	MU0DISCNT
	Revenue	XXXREV###
	Currency Key	0CURRENCY
	Sales Quantity	XXXQTY###
	Unit of Measure	0UNIT

13. Move the InfoObject Currency Key to **CUSTOMER**.
14. Move the InfoObject Unit of Measure to **PRODUCT**.
15. This is how your aDSO should look:

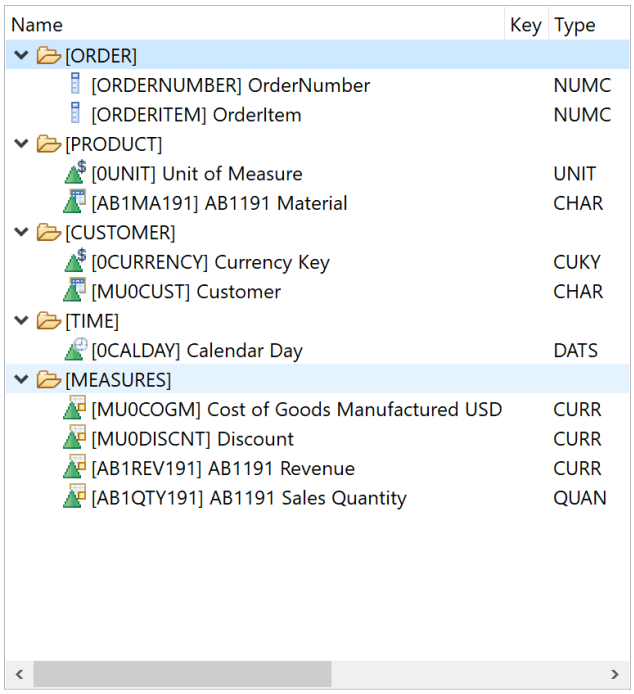


Figure 9

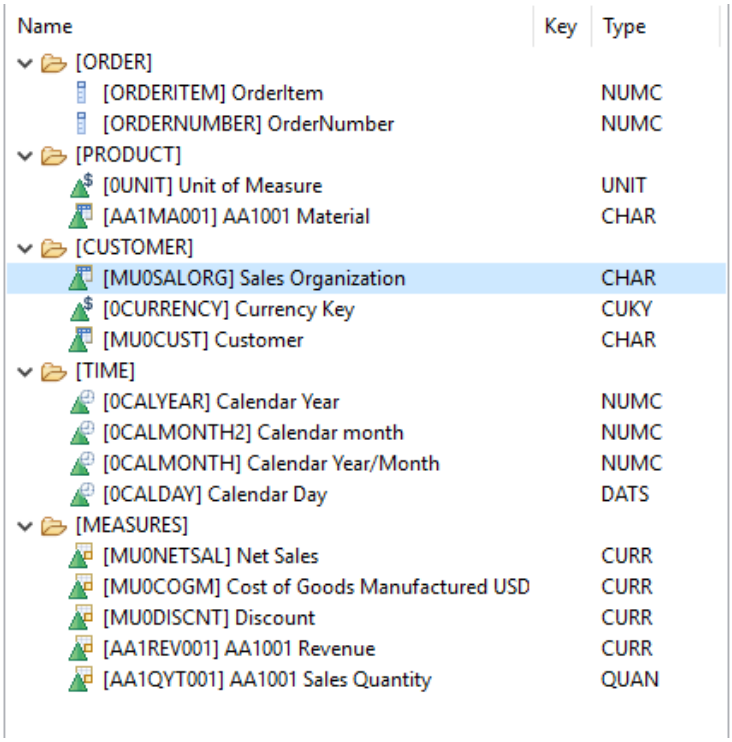
16. After this, remove all remaining Non-Key Fields (Currency, UnitOfMeasure) and then the folder Non-Key Fields.
17. Add more **InfoObjects** as the following characteristics and measures to your aDSO:

Group	InfoObject
MEASURES	Net Sales (delivered by UCC)
CUSTOMER	Sales Organization (delivered by UCC)
TIME	Calendar year (delivered by SAP)
TIME	Calendar month (delivered by SAP)
TIME	Calendar year/month (delivered by SAP)

Additional Information of handling of currency & unit: When you identify Key Figures like revenue or sales quantity, corresponding InfoObjects for currency (0CURRENCY) or unit of measurement (0BASEUOM) are added automatically by the system.

18. Move 0CURRENCY to group **CUSTOMER**, and 0UNIT to group **PRODUCT**, as they semantically belong to these entities.

19. Your aDSO should look like this (replace with appropriate InfoObjects that you created in previous exercise). Pay attention to the icons and technical names. *Correct any errors now otherwise your ETL process will fail in later steps!*



Name	Key	Type
[ORDER]		
[ORDERITEM] OrderItem		NUMC
[ORDERNUMBER] OrderNumber		NUMC
[PRODUCT]		
[0UNIT] Unit of Measure		UNIT
[AA1MA001] AA1001 Material		CHAR
[CUSTOMER]		
[MU0SALORG] Sales Organization		CHAR
[0CURRENCY] Currency Key		CUKY
[MU0CUST] Customer		CHAR
[TIME]		
[0CALYEAR] Calendar Year		NUMC
[0CALMONTH2] Calendar month		NUMC
[0CALMONTH] Calendar Year/Month		NUMC
[0CALDAY] Calendar Day		DATS
[MEASURES]		
[MU0NETSAL] Net Sales		CURR
[MU0COGM] Cost of Goods Manufactured USD		CURR
[MU0DISCNT] Discount		CURR
[AA1REV001] AA1001 Revenue		CURR
[AA1QYT001] AA1001 Sales Quantity		QUAN

Figure 10

20. Finally, **save** and **activate** your aDSO. If everything is correct, you get the following message:

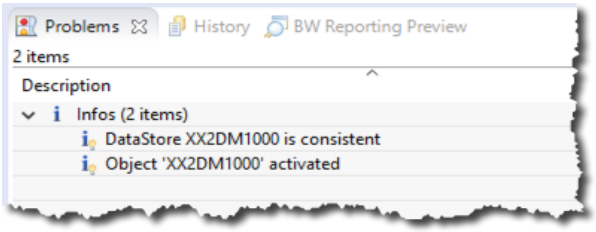


Figure 11

We have now modelled our main persistency, the semantically enriched data mart for worldwide sales data of Global Bike. As a new paradigm in-memory computing with SAP

BW/4HANA, all data are persisted on a detailed level and all aggregation is done on the fly. There is now multidimensional storage the advanced DataStore Object.

21. **Close** the aDSO.

Question 1: What is the difference between a field and an InfoObject?

Question 2: What is the advantage of using InfoObjects instead of fields in an aDSO?

3. Create a data flow object.

Create a data flow object to model the loading of sales transaction data for Global Bike Germany and US. Add the *Sales Transaction Data DataSource* from Global Bike Germany and US. Then, add the aDSO to the data flow.

1. Open **Eclipse**
2. Navigate to your InfoArea folder: XX → XXX → XXX### (name: Student ###).
3. Choose **New** → **Data Flow Object** in the context menu (right-click).
4. For the technical name: XXXDFSD### and as description XXX### Data Flow Sales Data.
5. **Finish**.
6. Drag and drop the **DataSource** icon (from the right panel **BW Object Types**) into your Data Flow.
7. To insert your DataSource choose **Add Object** in the context menu of the object that you just added.
8. Search and select your German Sales Data DataSource with the technical name XXXDSDE###.
9. Follow the steps above to also add your US Sales data datasource with the technical name XXXDSUS###.
10. Finally, add your aDSO XXXDM1### from previous step to the data flow.
11. After finishing this, your Data Flow should look like the picture below.

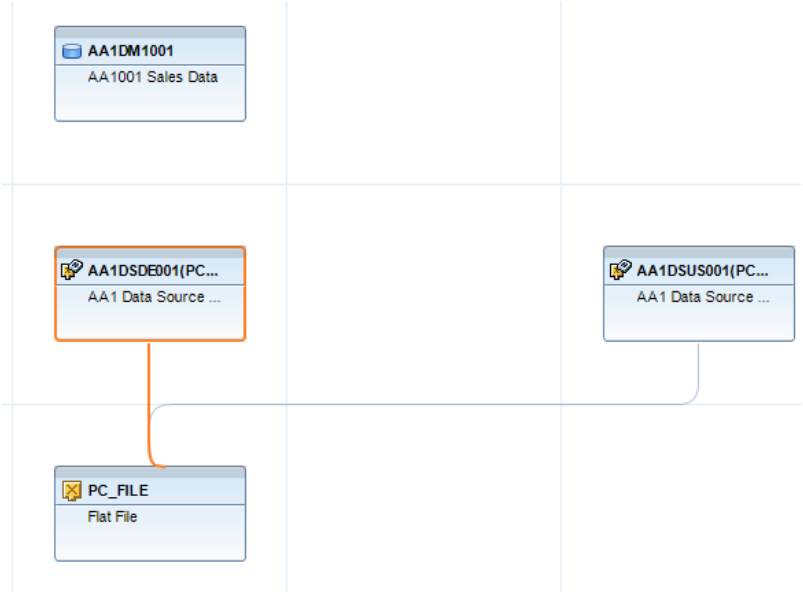


Figure 12

12. **Save** and **activate** the data flow.

Question 3: Insert a screenshot of your Data Flow and label each box with the *type* of object it is (not the name).

4. Create and define rules for a transformation.

Now you will define the *transformation* rules between the data acquisition layer (sales transaction Data Source from Global Bike Germany and US) and the data mart layer (advanced DataStore Object).

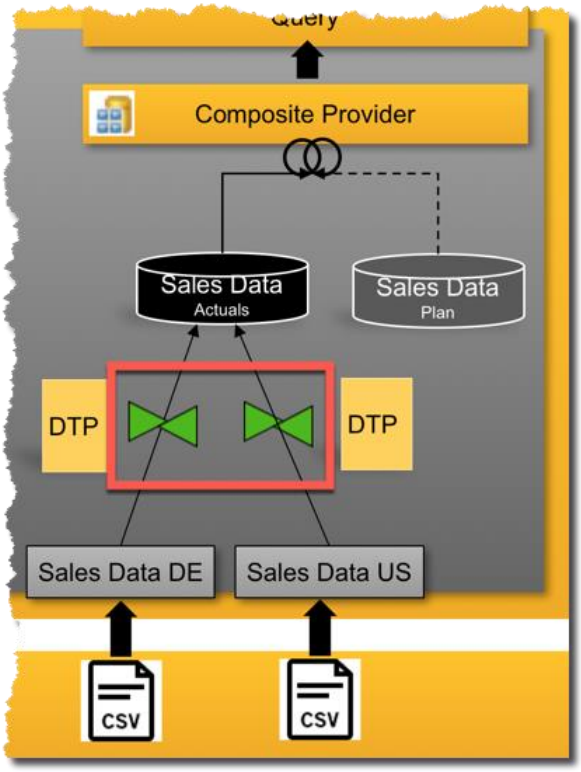


Figure 13

1. Open **Eclipse**
2. Navigate to your InfoArea folder: XX → XXX → XXX### (name: Student ###).
3. Open your Data Flow from previous step with technical name XXXDFSD###.
4. **Connect** the left DataSource (Sales Data DE ###) with the advanced DataStore Object (XXX### Sales Data) by drawing a line between the two objects using the arrow.

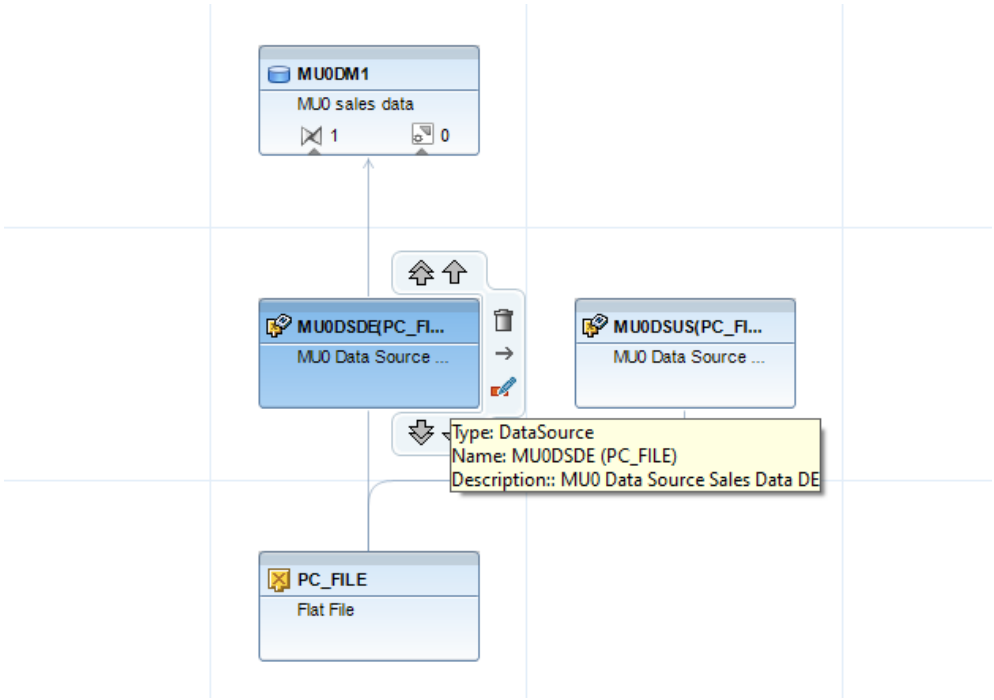
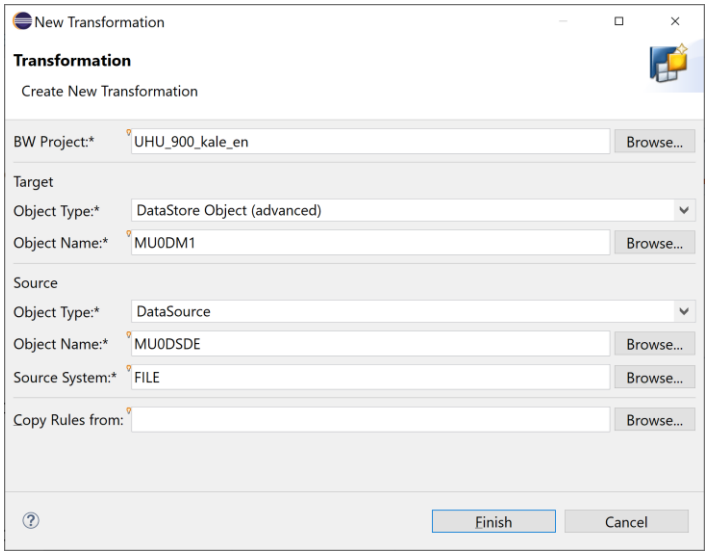


Figure 14

5. **Save** the DataFlow.
6. Create a transformation via double-click on the white transformation symbol.



7. **Finish**.
8. Click on **Rules** Tab.
9. In the transformation editor, define the transformation rules by mapping the source and target fields. As compared to the transformation of the material master data, we

have in this case not only direct assignment of field/InfoObjects, but in addition more complex rules.

10. First, assign the fields of the DataSource with the fields/InfoObjects in the advanced DataStore Object that can be *directly assigned*.
    a. **CUSTOMER** to **MU0CUST** (description: Customer)
    b. **PRODUCT** to **XXXMA###** (description: XXX### Material)
    c. **REVENUE** to **XXXREV###** (description: XXX### Revenue)
    d. **SALESQUANTITY** to **XXXQTY###** (description: XXX### Sales Quantity)
    e. **DISCOUNT** to **MU0DISCNT** (description: Discount)
    f. **COSTSINUSD** to **MU0COGM** (description: Cost of Goods Manufactured USD)

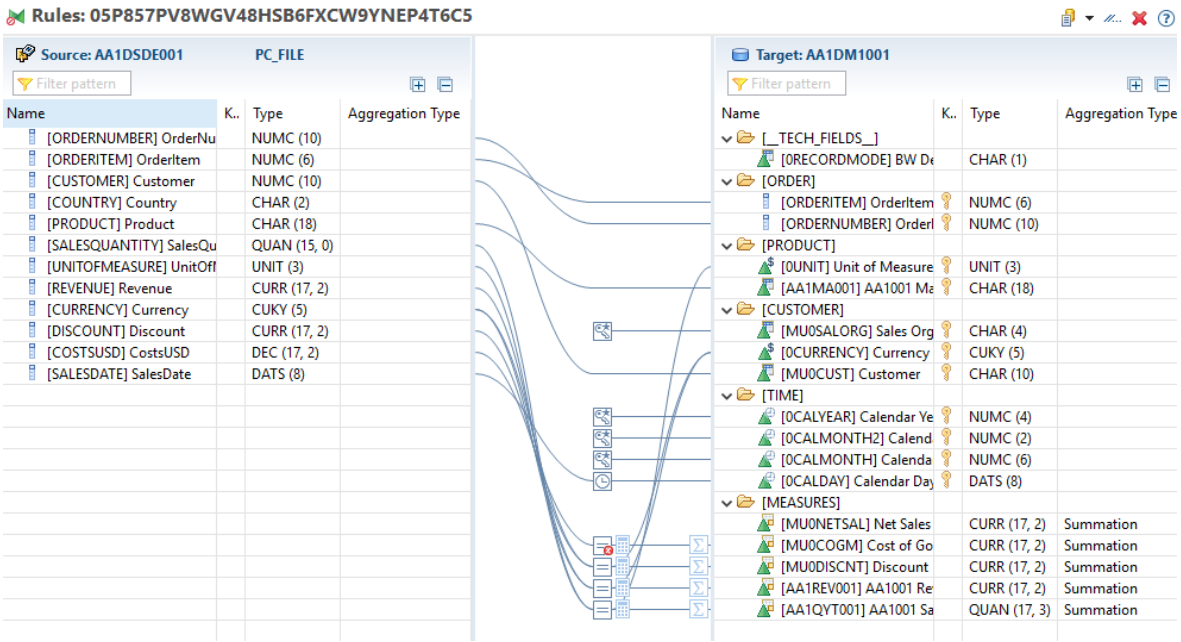


Figure 15

11. The key figure Cost of Goods Manufactured USD has fixed currency USD. Double-click the = sign for Cost of Goods Manufactured.
12. Since our source field is in USD as well we have to choose **Direct Assignment** in the rule.
13. **Uncheck** Conversion Type

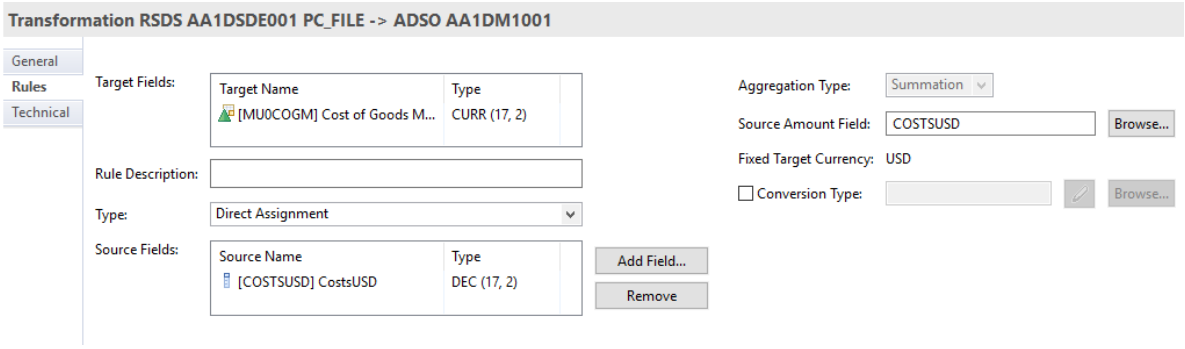


Figure 16

14. **Save.**

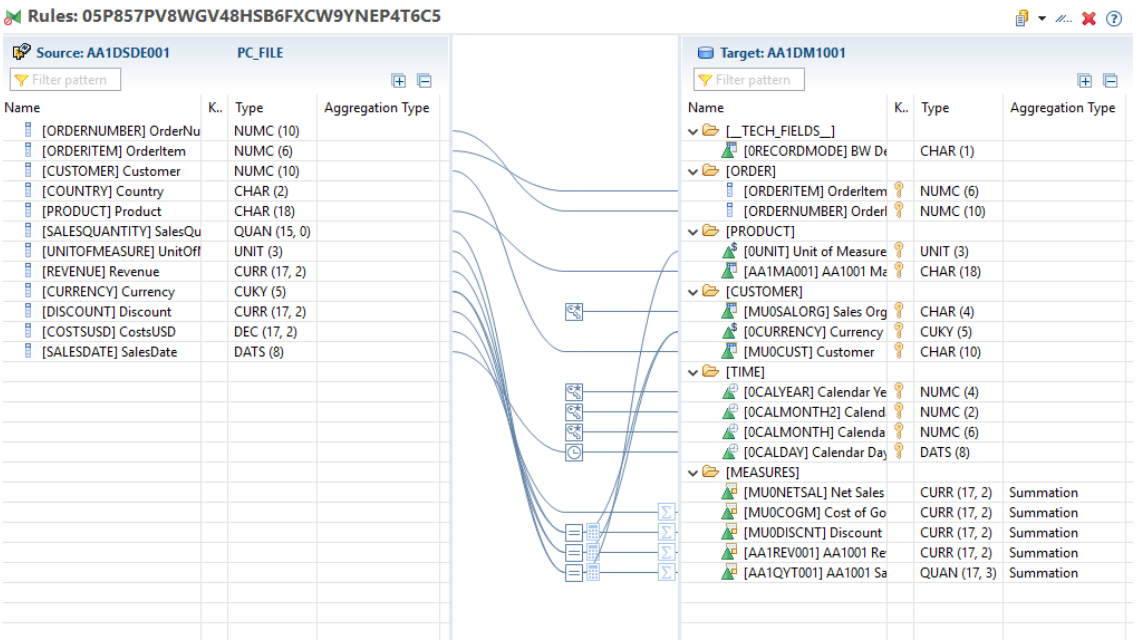


Figure 17

15. The system can derive all *time characteristics* automatically from SalesDate as source field. This contains oCALYEAR (Calendar Year), OCALMONTH2 (Calendar month), oCALMONTH (Calendar Year/Month), oCALDAY (Calendar Day).

16. **Connect** all of the source and target objects as shown below.

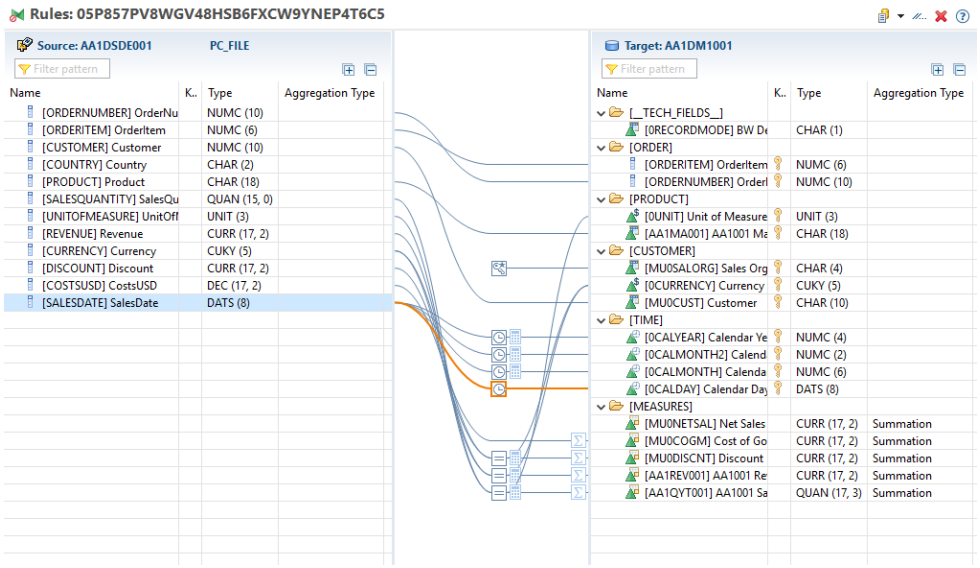


Figure 18

17. Configure the transformation of Calendar Day as follows:

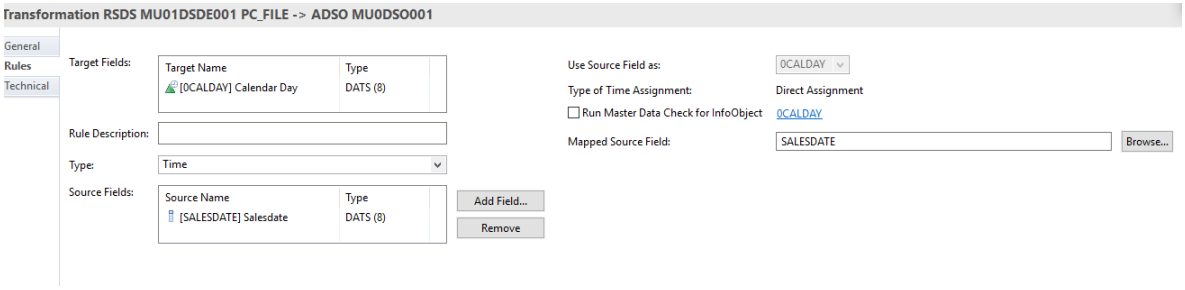


Figure 19

18. **Save.**

19. Configure Net Sales which is defined as a rule of type "Formula":

Net Sales = Revenue - Discount

20. To define this formula, add REVENUE and DISCOUNT as source fields for the rule (MU0NETSA, Net Sales), then click on the target field Net Sales and choose then rule type Formula. The net sales is calculated during load process, record by record. Set the options as shown below.

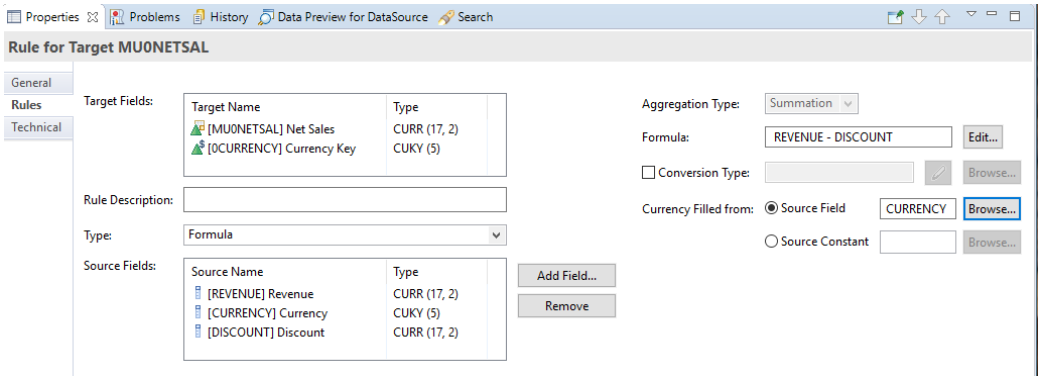


Figure 20

21. **Save**.
22. There is only one InfoObject that does not have a transformation yet.
23. Sales Organization is not part of transaction data but in the master data table of the InfoObject Customer (Customer has an attribute Sales Organization).
24. So, the idea is to do a "*master data lookup*" when loading the data to the data mart layer.
25. **Configure** Sales Organization. It is a time-dependent attribute of customer. Therefore, the system must know the date when reading the attribute.
26. Change the **Rule Type** to Lookup.
27. Add the Customer first.
28. The lookup object is Customer.
29. Set the following options.

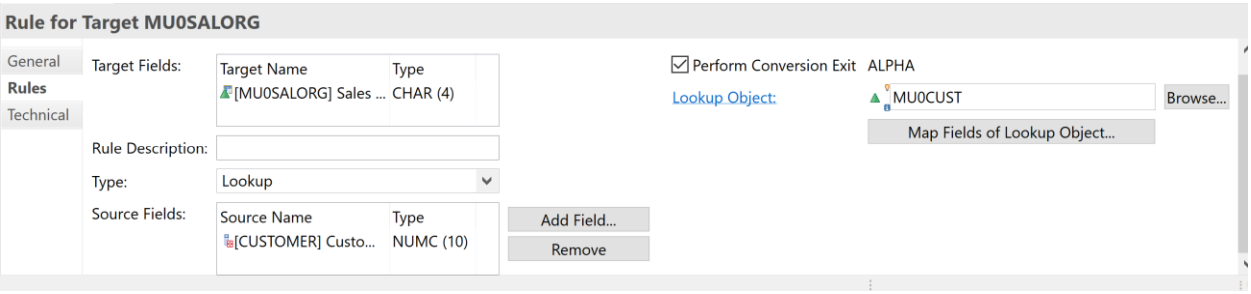


Figure 21

30. Then the Salesdate field will be added automatically in the next step.
31. Click on **Map Fields for Lookup Object**. Fill all of the fields as shown below.

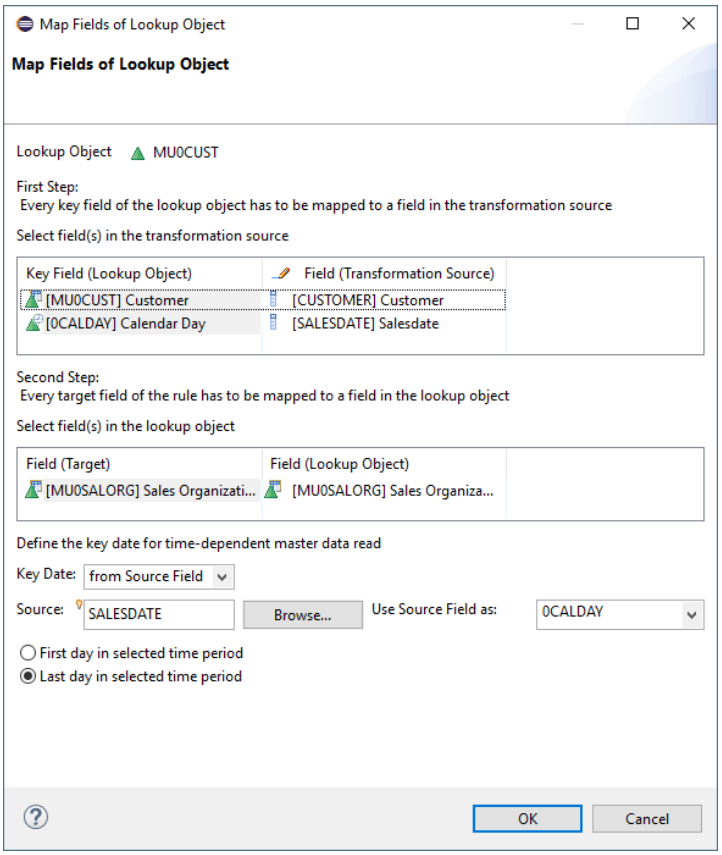


Figure 22

32. Click OK.

33. Your rule should look like this.

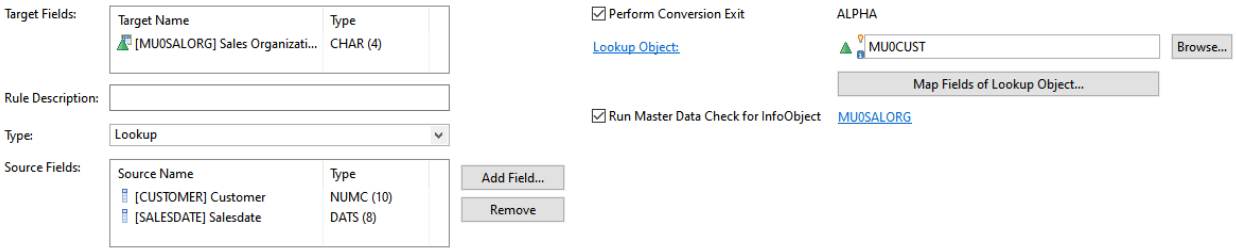


Figure 23

34. **Save** and **Activate** your transformation (warning can be ignored).

35. **Close** the transformation.

36. Go back to your data flow and **save** and **activate** the data flow.

37. The transformation icon should be green (bowtie).

38. Connect the second **DataSource for US** with the **DataStore Object** - Right-click the **Sales Data ADSO → Transformation → Create Transformation**.

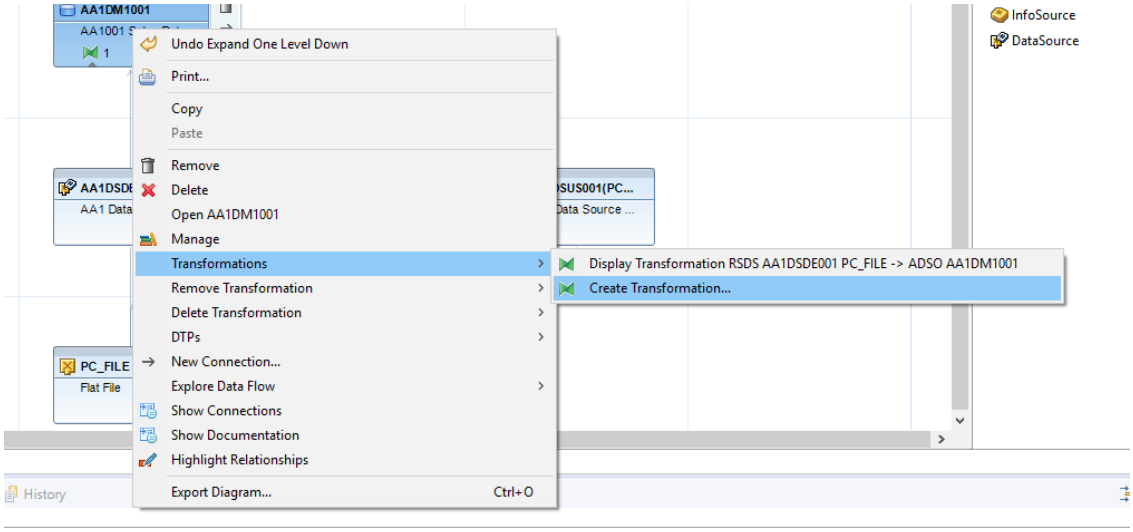


Figure 24

39. On the following popup, name of the DataSource in XXXDSUS###.

40. **Copy Rules** from your XXXDSDE### to XXXDM1### transformation.

41. **Finish**.

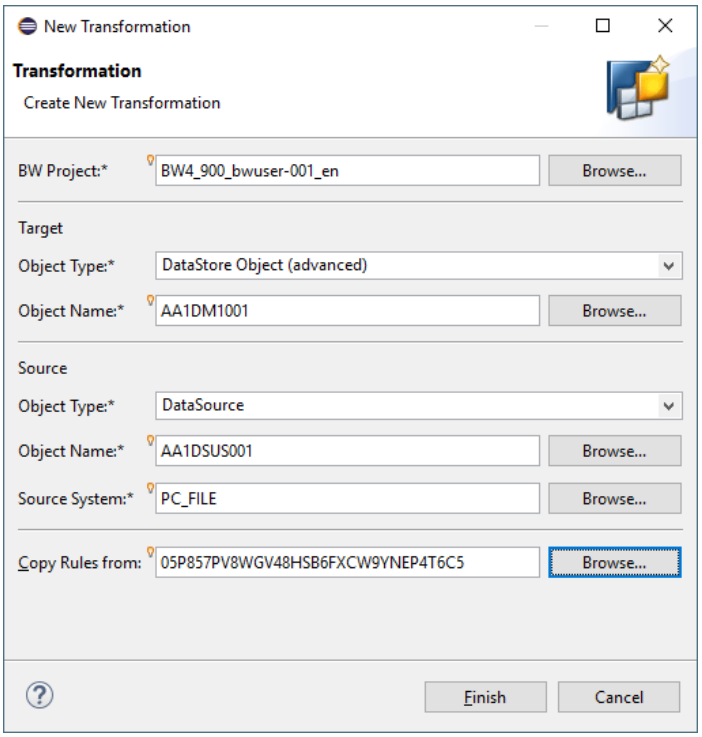


Figure 25

42. **Save** and **Activate** your transformation (warning can be ignored).

43. **Close** transformation.

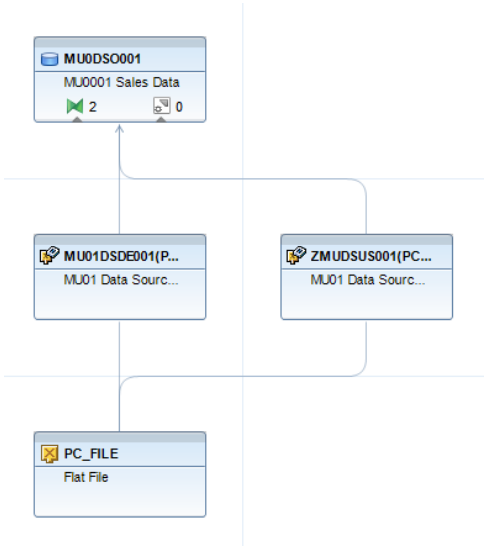


Figure 26

44. **Save** and **activate** the data flow.

Question 4: Why did we choose Direct Assignment for Customer?

Question 5: Can one source object be mapped to multiple target objects? Provide an example.

Question 6: Why does the Cost of Good Manufactured not need a currency key?

Question 7: What will be the final currency of Net Sales?

Question 8: Explain, in detail, the transformation rule for Sales Organization.

Question 9: Why does the Sales Data aDSO need two transformations?

Question 10: Why is the ordernumber a field and not an InfoObject?

5. Create a Data Transfer Process and load data.

Jan creates a *Data Transfer* Process between the data acquisition layer (sales transaction Data Source from Global Bike Germany and US) and the data mart layer (aDSO). Then, he loads the data into the data warehouse.

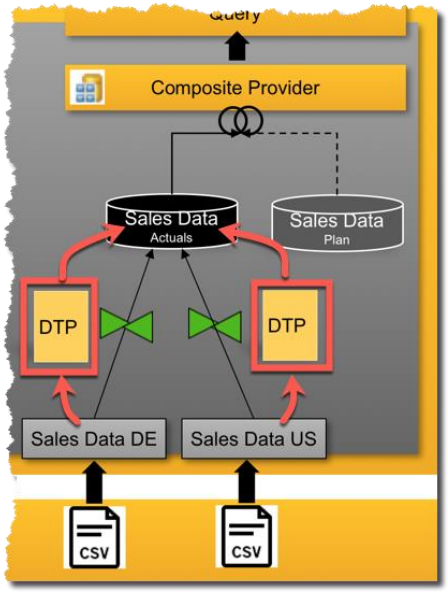


Figure 27

1. Open **Eclipse**.
2. Navigate to your InfoArea folder XX → XXX → XXX### (name: Student ###). Open your Data Flow from previous step with technical name XXXDFSD###.
3. Create a Data Transfer Process (DTP) between the left DataSource (Sales Data DE ###) and the aDataStore Object (XXX### Sales Data) (**draw a line** using the arrow).

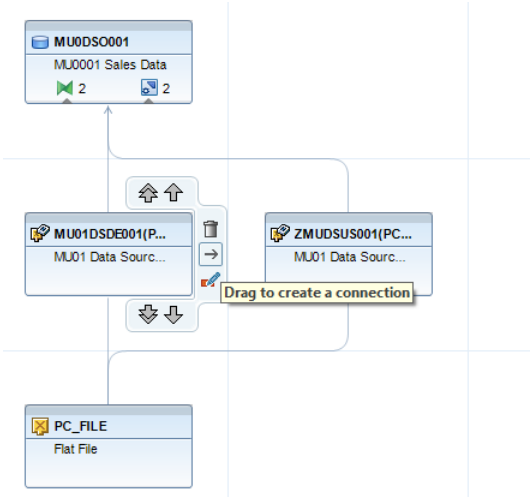


Figure 28

4. Double-click on **DTP**.
5. **Next** → **Finish**.
6. Configure the following settings:

a. Extraction Mode: Full

b. **Extraction** tab

(1) File Name: Browse for your *Sales data DE.csv* file.

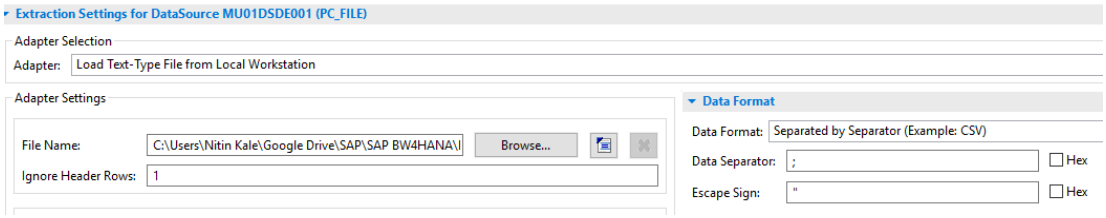


Figure 29

7. **Save** and **Activate** the DTP.

8. Click on Start Execution of DTP (top right).

9. Click on

.

10. **Enter → Allow → Yes.**

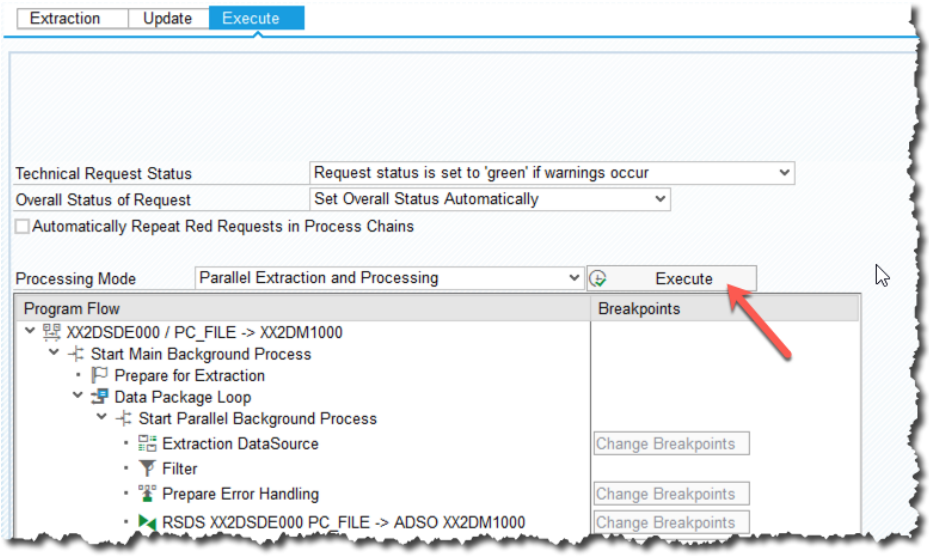


Figure 30

11. Check the monitor if everything was ok (refresh may be necessary).

12. Click through any popups. Be sure that 86,816 transactions have been loaded.

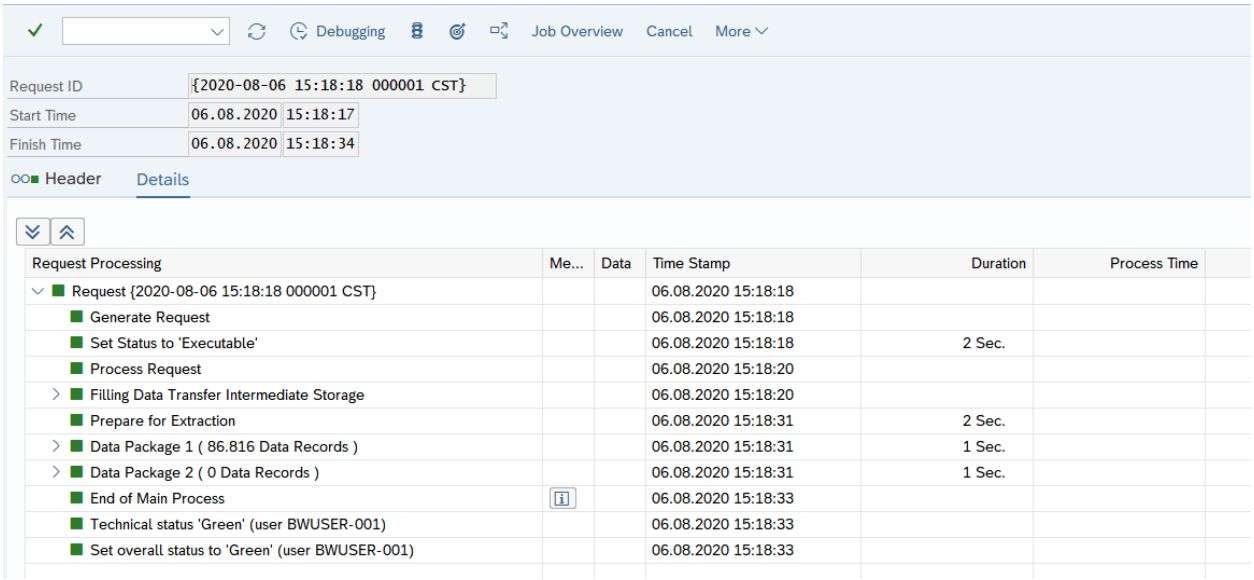


Figure 31

13. **Close** the **Execute** tab.
14. **Close** DTP.
15. Go back to your **Data Flow** and repeat the steps (on your own) for creating and executing the DTP for the second Data Source (on the right side) for US.
16. Remember to change the file in the **Extraction** tab to *Sales Data US.csv*.
17. 84,194 transactions for the US should have been loaded.
18. After you have completed the DTP for US, go back to Data Flow and **save** it.
19. If data are loaded successfully into the aDataStore Object, they can be **activated**.
20. Go to the BW cockpit URL provided by your UCC
21. Go to Data Management, then click on DataStore Objects Manage Requests

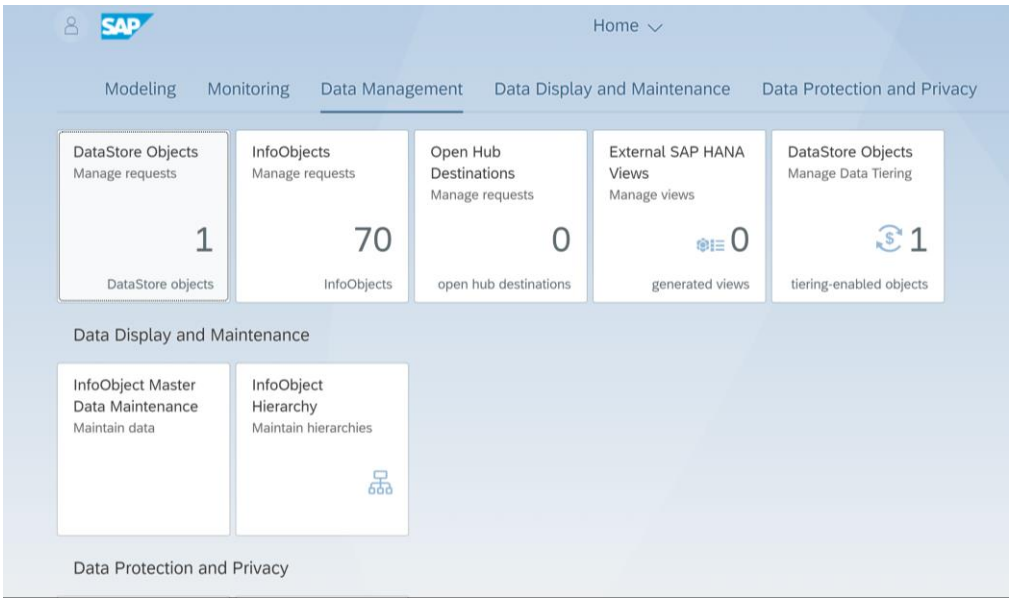


Figure 32

22. On the following screen, you see an overview over all requests loaded into your aDSO. The requests are grouped by day when you executed the DTP.

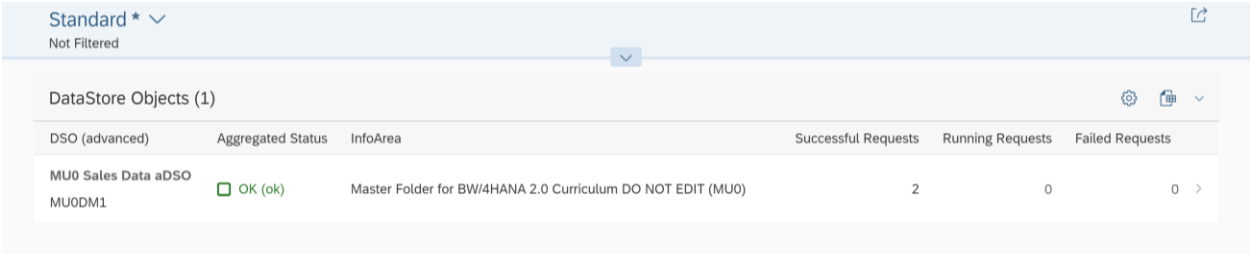


Figure 33

23. Click your aDSO
24. Select the **Request** and then **Activate**.

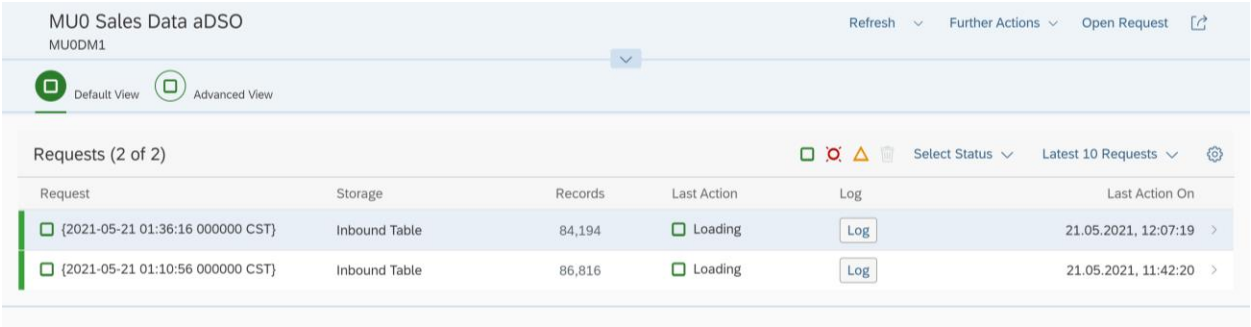


Figure 34

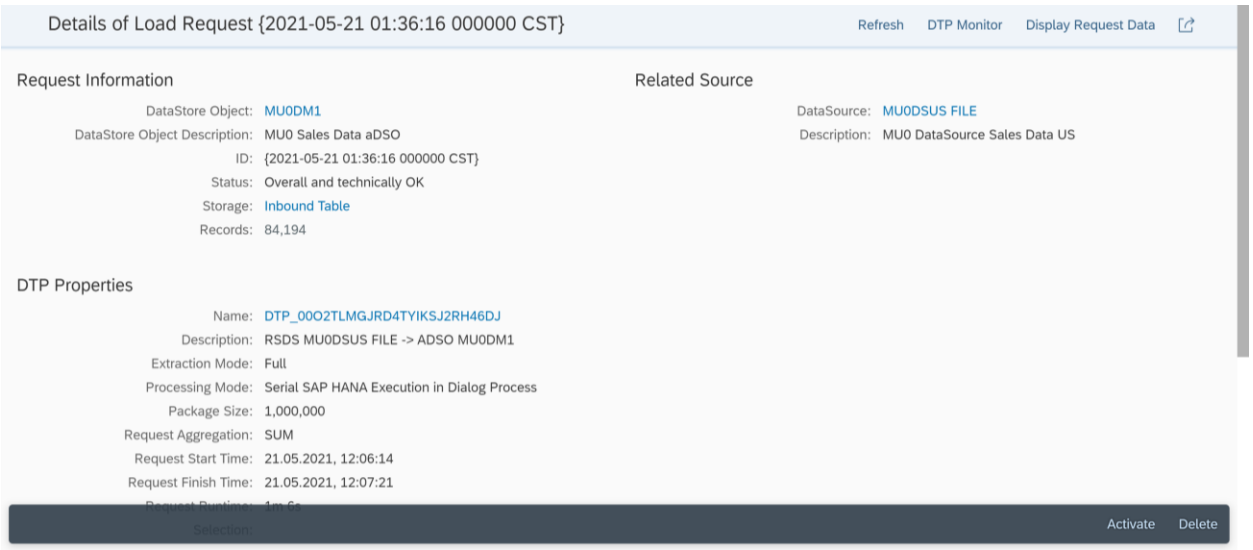


Figure 35

25. After successful activation, the request overview looks like this. You may need to Refresh.

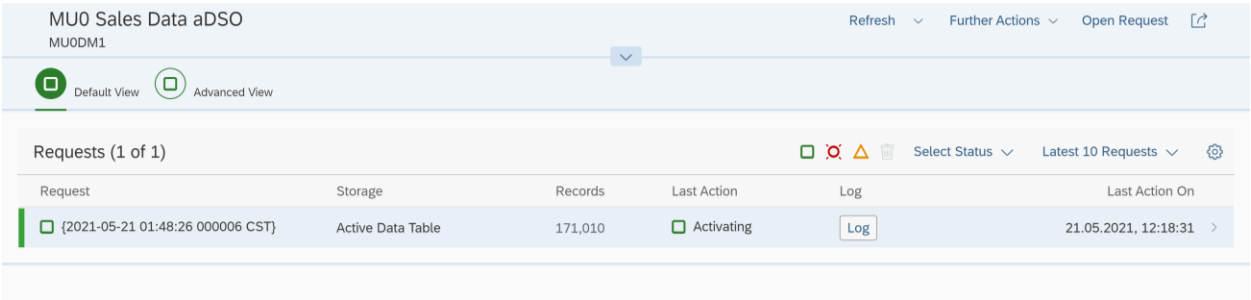


Figure 36

26. Your data is now loaded and can be used for analytics from the aDSO. Go back to your Data Flow (in Eclipse). **Save** and **activate** it.

27. **Close** the Data Flow Object. In the ETL process what does DTP represent?

28. Open your aDSO in Eclipse. Click on Data Preview. You should confirm that the total Cost of Goods Manufactured is $953,354,225.28

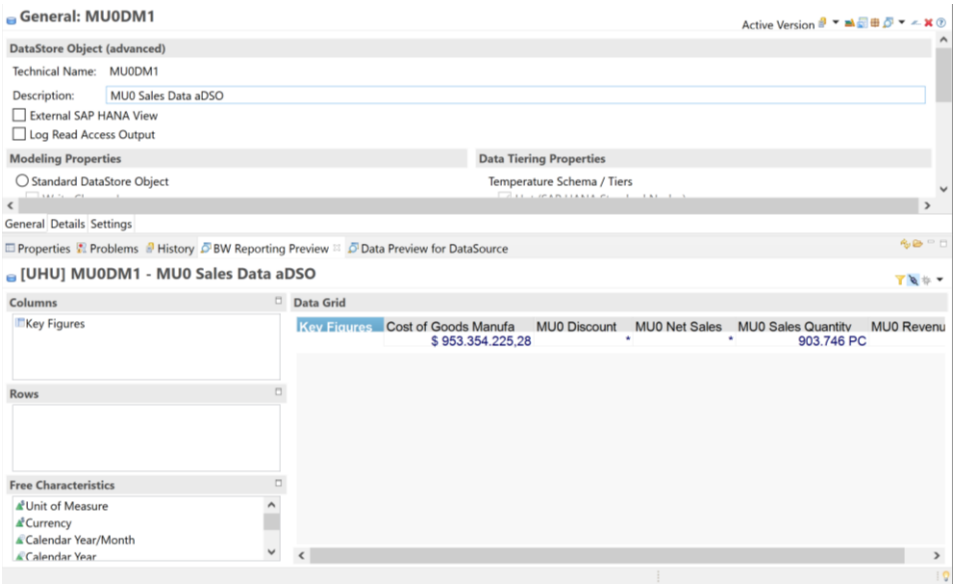


Figure 37

Question 11: Why did you have to execute two DTPs?

Question 12: What would happen if you executed a DTP twice by mistake?

Question 13: What is the source object of the DTP?

Question 14: What is the target object of the DTP?

Question 15: In these DTP, what type of data are we loading (in the context of the snowflake schema)?

6. Create a CompositeProvider.

Jan uses Eclipse to create the virtualization layer of the data warehouse with a *CompositeProvider*.

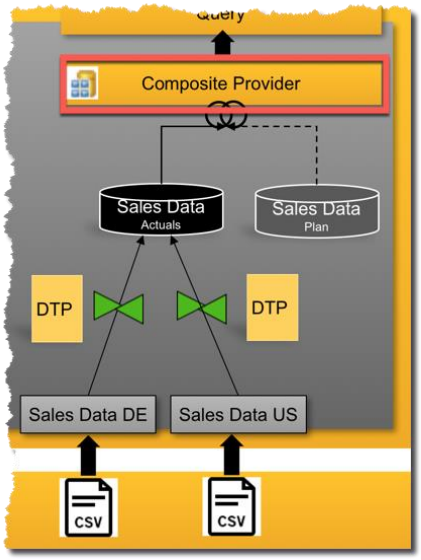


Figure 38

1. Start **Eclipse** and open your BW project: **Project Explorer → your BW-System → BW Repository**.
2. Open your BW project in Eclipse and navigate to the InfoArea assigned by your instructor, which has the technical name XXX### and description Student ###.
3. Choose **New → CompositeProvider**.

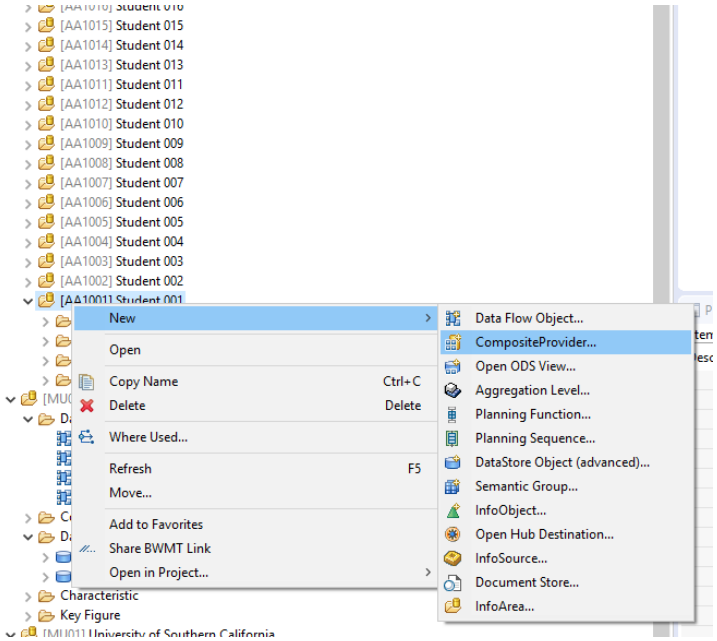


Figure 39

4. On the following screen, choose as name XXXCP1### and as description XXX### Sales Data Composite Provider.

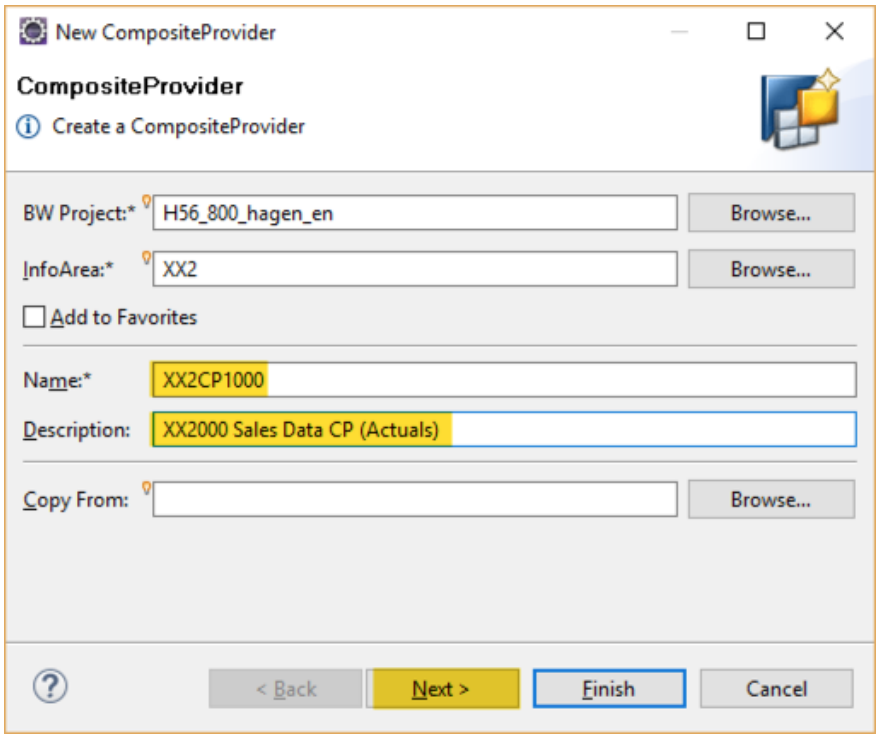
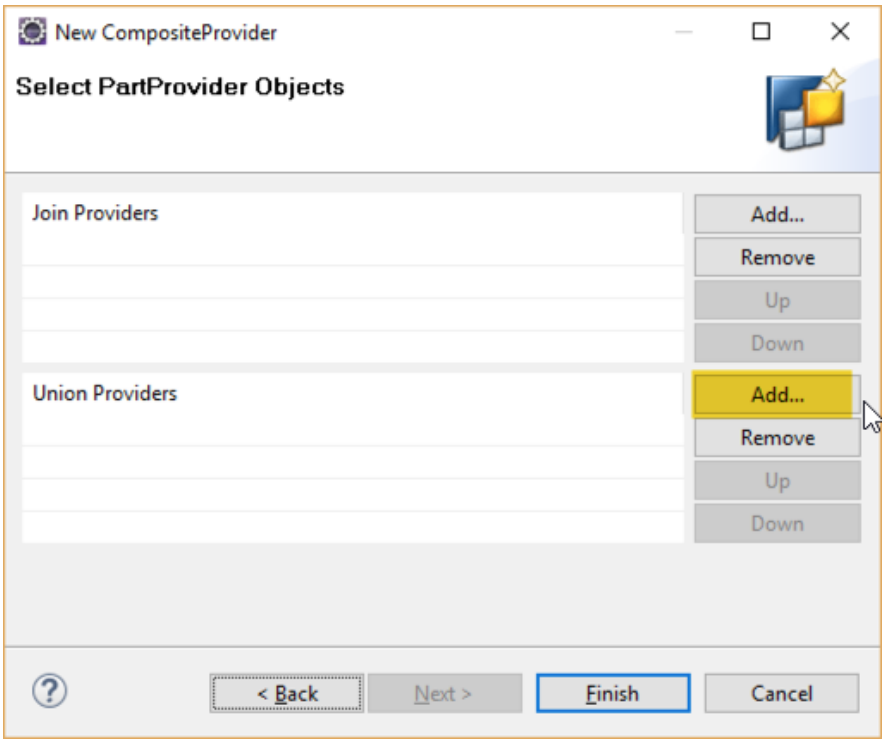


Figure 40

5. Choose **Next** to add your aDataStore Object (technical name: XXXDM1###) as part provider in the **Union** section.



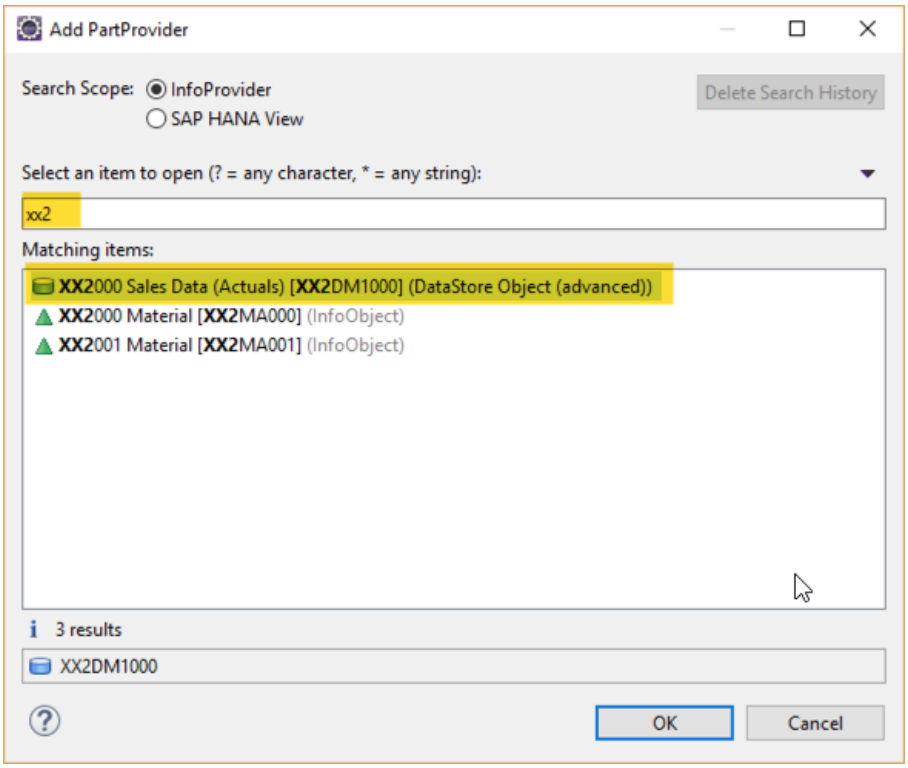


Figure 41

6. Choose **Finish** to create the CompositeProvider (CP) and open the editor for this object. Save.

7. Extend the CompositeProvider

Jan uses Eclipse to define *join* and *union conditions* to link the part providers that make the CompositeProvider.

1. Start **Eclipse** and open your BW project: **Project Explorer → your BW-System → BW Repository**.
2. Open your CompositeProvider from previous step with technical name XXXCP1###.
3. On the tab **Scenario**, you define in general the join and union conditions to link the part providers that make the CompositeProvider.

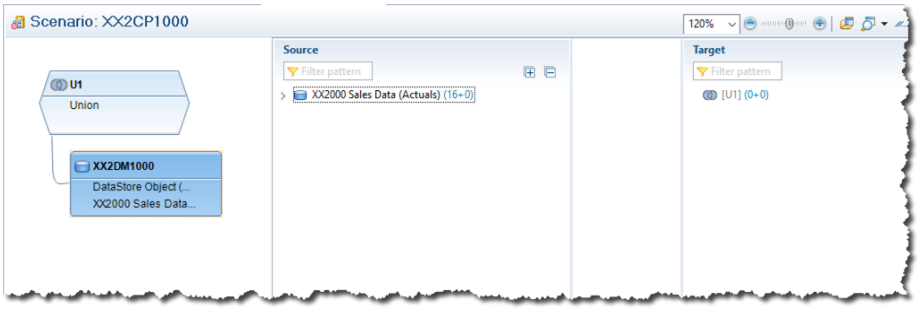


Figure 42

4. In our case, we have only one InfoProvider.
5. Just **drag** from source to target to assign the fields to the target structure (a block, not field by field).

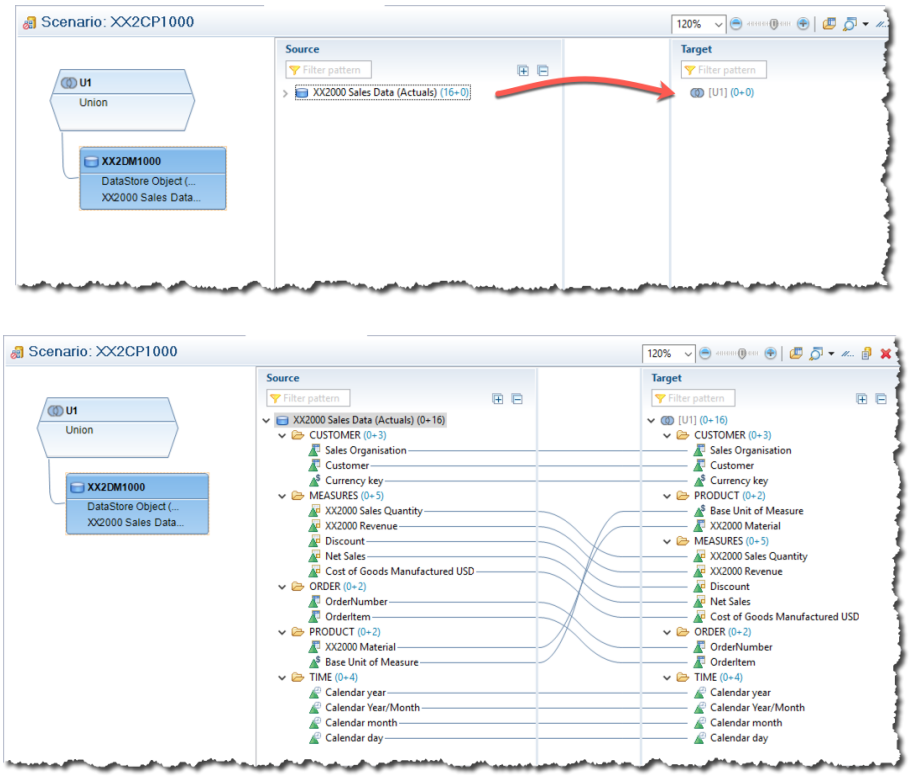


Figure 43

6. **Save** and **activate** your CompositeProvider.

Question 16: What is the difference between an aDSO and a composite provider?

Question 17: Is the data in a composite provider persistent?

8. Change properties of the output structure of the CompositeProvider.

Jan uses Eclipse to extend the CompositeProvider. For this, he changes some *properties* of the *output* structure of the CompositeProvider.

1. Start **Eclipse** and open your BW project: **Project Explorer → your BW-System → BW Repository**.
2. Open your CompositeProvider from previous step with technical name XXXCP1###.
3. On tab **Output** we can change some properties of the output structure of the CP, e.g. names and we can enrich the CP with navigation attributes of the underlying InfoObjects.

4. First, change the description of InfoObject XXXMA### to Product, as this is the term the sales department would prefer to use.

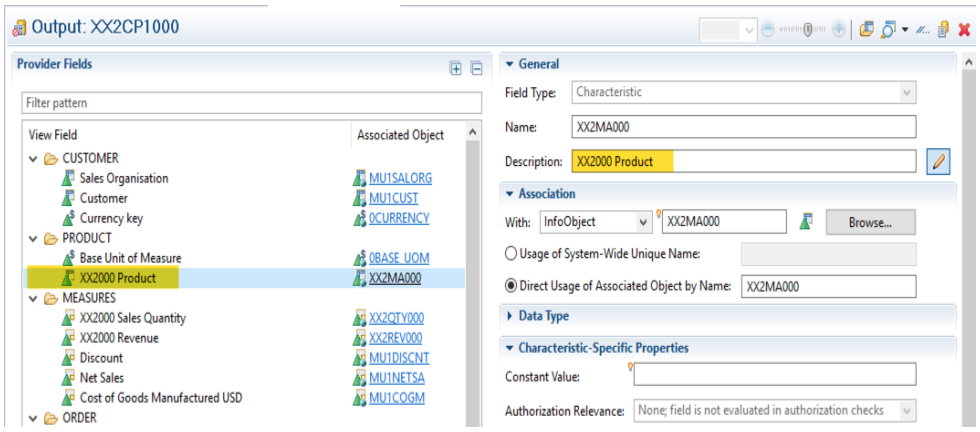
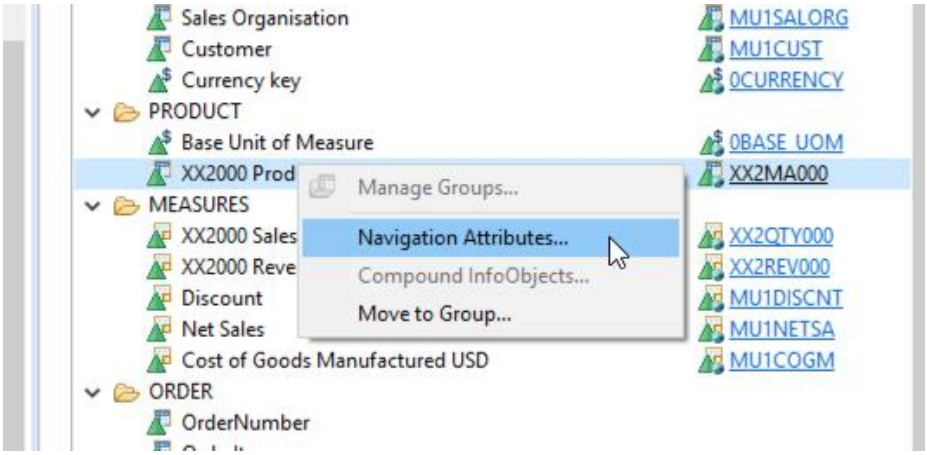


Figure 44

5. In the context menu (right-click) of InfoObject Product (XXXMA###) choose **Navigation Attributes**.

6. Switch on the navigation attributes Product Category (XXXPC###) and Division (MU0DIVN).



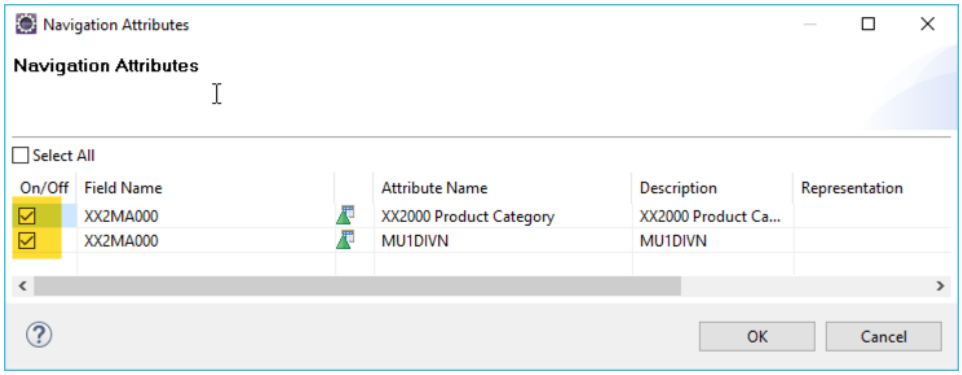
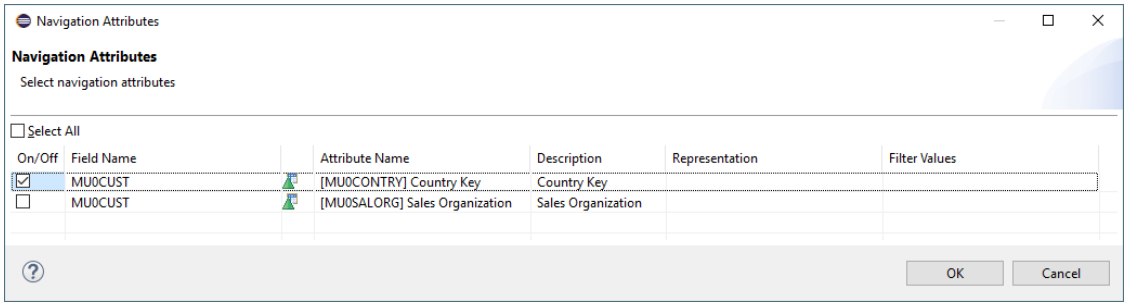


Figure 45

7. In the context menu of InfoObject Customer (MU0CUST) choose **Navigation Attributes**.
8. Switch on the navigation attribute Country (MU0COUNTRY).



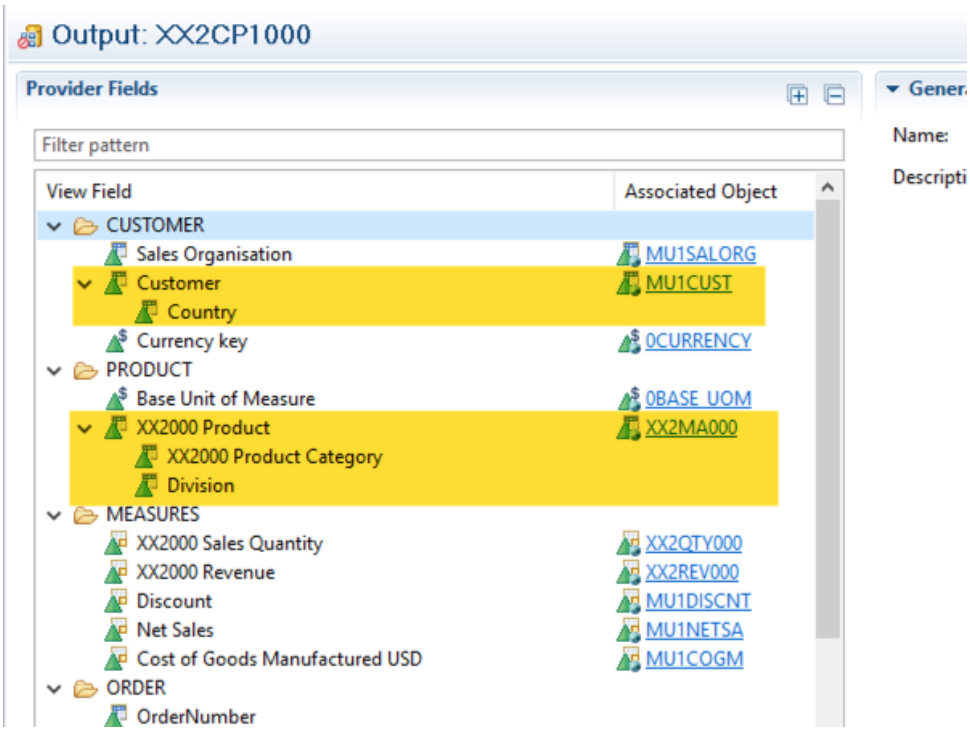


Figure 46

9. Finally, **save** and **activate** your CompositeProvider.
10. In case of success, you get this message:

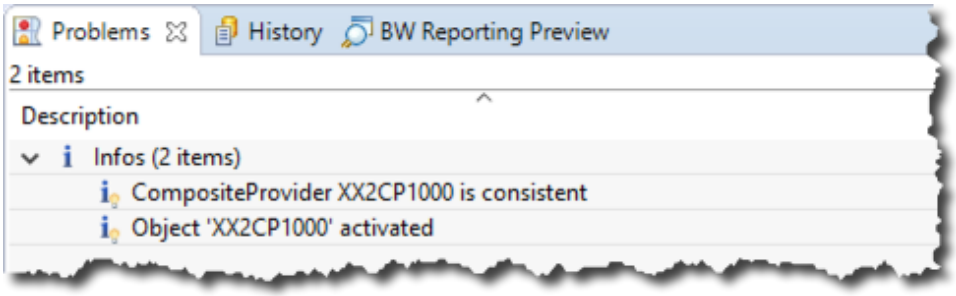


Figure 47

11. You can test your CP with the "Data Preview". Click on the upper right side on and choose **Data Preview**.

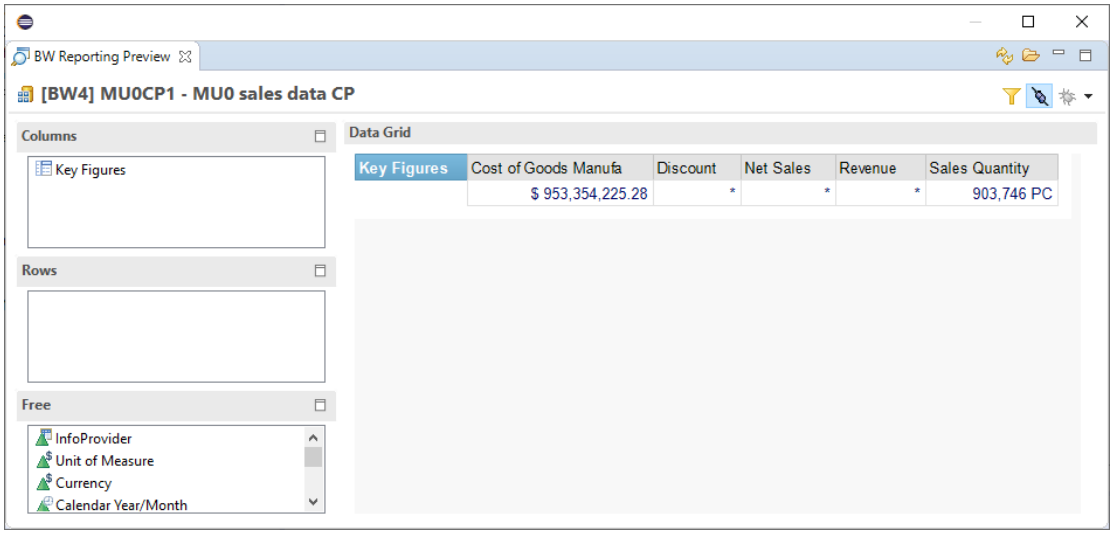


Figure 48

12. Without any drill down:

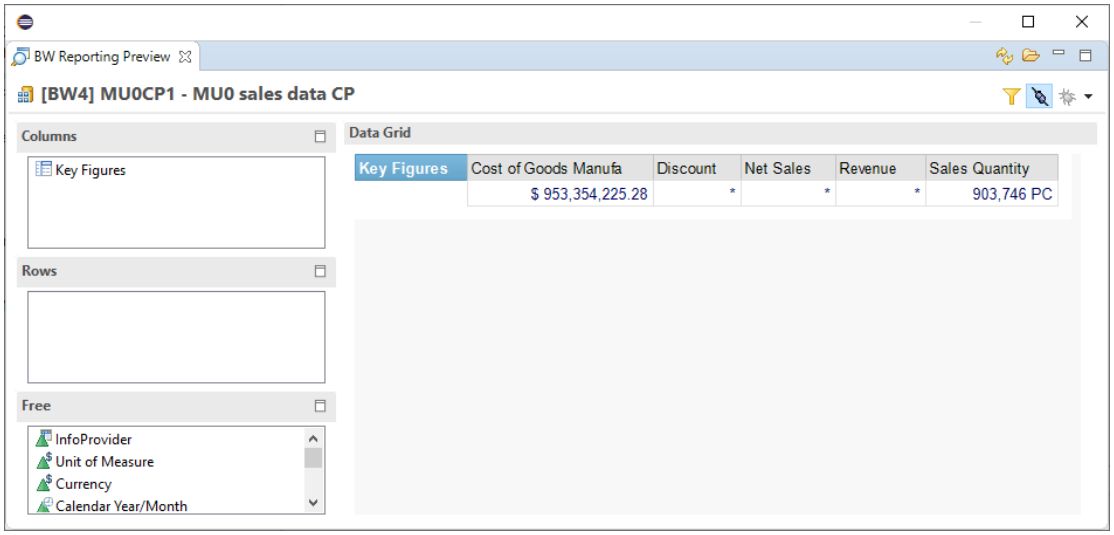
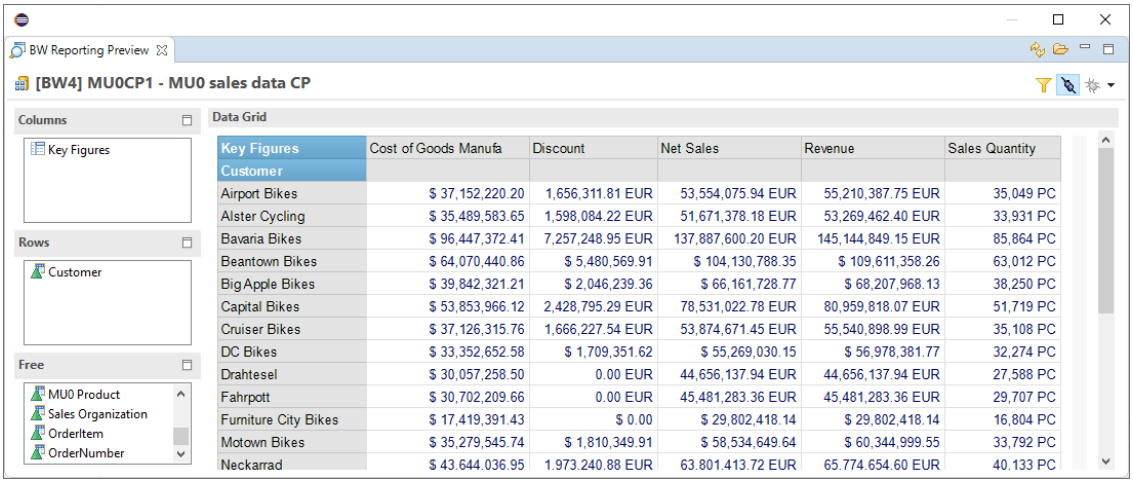


Figure 49

13. Drill down by customer: **Drag and drop** Customer into **Rows**.



Key Figures	Cost of Goods Manufa	Discount	Net Sales	Revenue	Sales Quantity
Customer					
Airport Bikes	$ 37,152,220.20	1,656,311.81 EUR	53,554,075.94 EUR	55,210,387.75 EUR	35,049 PC
Alster Cycling	$ 35,489,583.65	1,598,084.22 EUR	51,671,378.18 EUR	53,269,462.40 EUR	33,931 PC
Bavaria Bikes	$ 96,447,372.41	7,257,248.95 EUR	137,887,600.20 EUR	145,144,849.15 EUR	85,864 PC
Beantown Bikes	$ 64,070,440.86	$ 5,480,569.91	$ 104,130,788.35	$ 109,611,358.26	63,012 PC
Big Apple Bikes	$ 39,842,321.21	$ 2,046,239.36	$ 66,161,728.77	$ 68,207,968.13	38,250 PC
Capital Bikes	$ 53,853,966.12	2,428,795.29 EUR	78,531,022.78 EUR	80,959,818.07 EUR	51,719 PC
Cruiser Bikes	$ 37,126,315.76	1,666,227.54 EUR	53,874,671.45 EUR	55,540,898.99 EUR	35,108 PC
DC Bikes	$ 33,352,652.58	$ 1,709,351.62	$ 55,269,030.15	$ 56,978,381.77	32,274 PC
Drahtesel	$ 30,057,258.50	0.00 EUR	44,656,137.94 EUR	44,656,137.94 EUR	27,588 PC
Fahrpott	$ 30,702,209.66	0.00 EUR	45,481,283.36 EUR	45,481,283.36 EUR	29,707 PC
Furniture City Bikes	$ 17,419,391.43	$ 0.00	$ 29,802,418.14	$ 29,802,418.14	16,804 PC
Motown Bikes	$ 35,279,545.74	$ 1,810,349.91	$ 58,534,649.64	$ 60,344,999.55	33,792 PC
Neckarrad	$ 43.644.036.95	1.973.240.88 EUR	63.801.413.72 EUR	65.774.654.60 EUR	40.133 PC

Figure 50

14. You have now modeled, designed, implemented, and populated SAP BW/4HANA 2.0 Data Warehouse! It is now ready for analytics!

Question 18: What object in all of SAP BW/4HANA represents the snowflake schema?

Question 19: List the process (at least 10 steps) of building and populating a BW/4HANA 2.0 data warehouse. Refer to all previous exercises in Chapters 3 and 4.

Question 20: Now that you have modeled, built and populated a data warehouse in SAP BW/4HANA 2.0, list one advantage to snowflake schemas that we had discussed in class but did not implement in our data warehouse.